



SAPIENZA
UNIVERSITÀ DI ROMA



Proactive Defense against Malicious Generative AI

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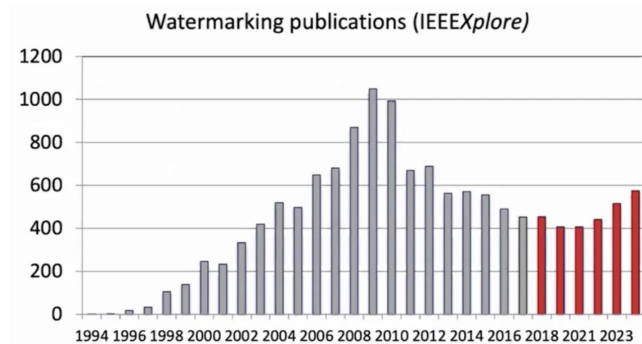


OmnAI Lab



Tracing AI-Generated Content

- Generative AI is everywhere and anyone can access to it at no cost.
- It can be used for good...and for bad purposes.
- Watermarking: Embed some hidden information inside a multimedia content for having a claim.
- Alternatives to watermarks include immunizations, digital signatures, etc.



Some Examples of Protections

SynthID  Gemini

- SynthID (inside media)
- Properties:
 - Non-informative
 - Deep learning-based
 - No Key
 - Content Watermarking
 - Multimedia
 - Non-cryptographic
 - Proprietary



Coalition for
Content Provenance
and Authenticity

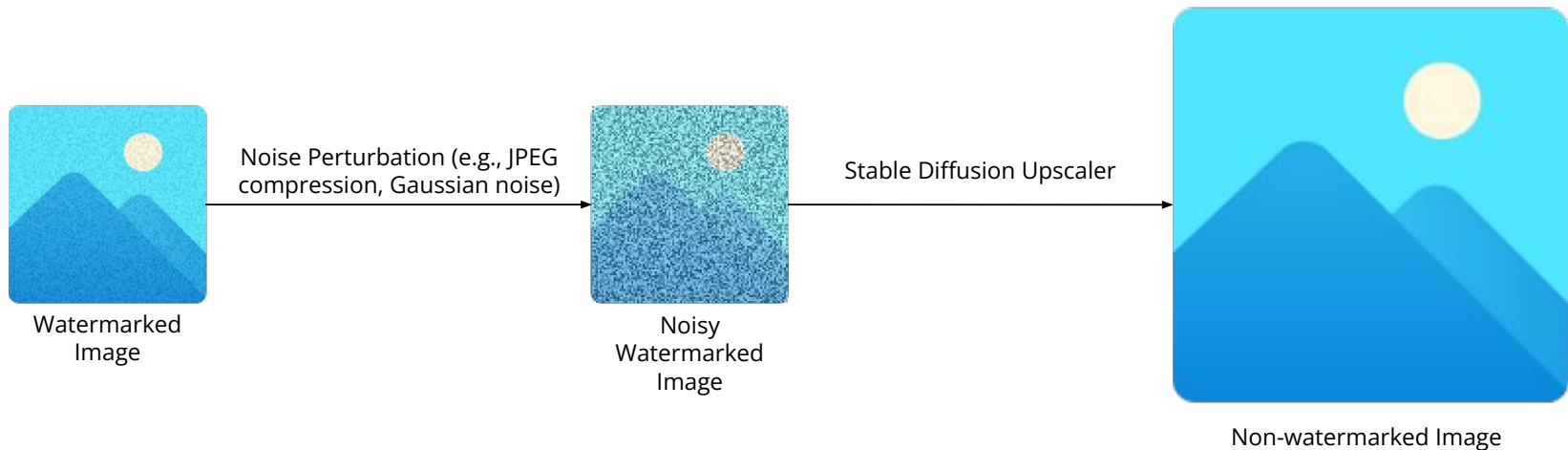
- C2PA (metadata)
- Properties:
 - Informative
 - Steganographic
 - Secret Key
 - Content Watermarking
 - Metadata
 - Cryptographic
 - Open

 Both methods are vulnerable!

Gowal, S., Bunel, R., Stimberg, F., Stutz, D., Ortiz-Jimenez, G., Kouridi, C., et al. (2025). SynthID-Image: Image watermarking at internet scale. *arXiv preprint arXiv:2510.09263*.

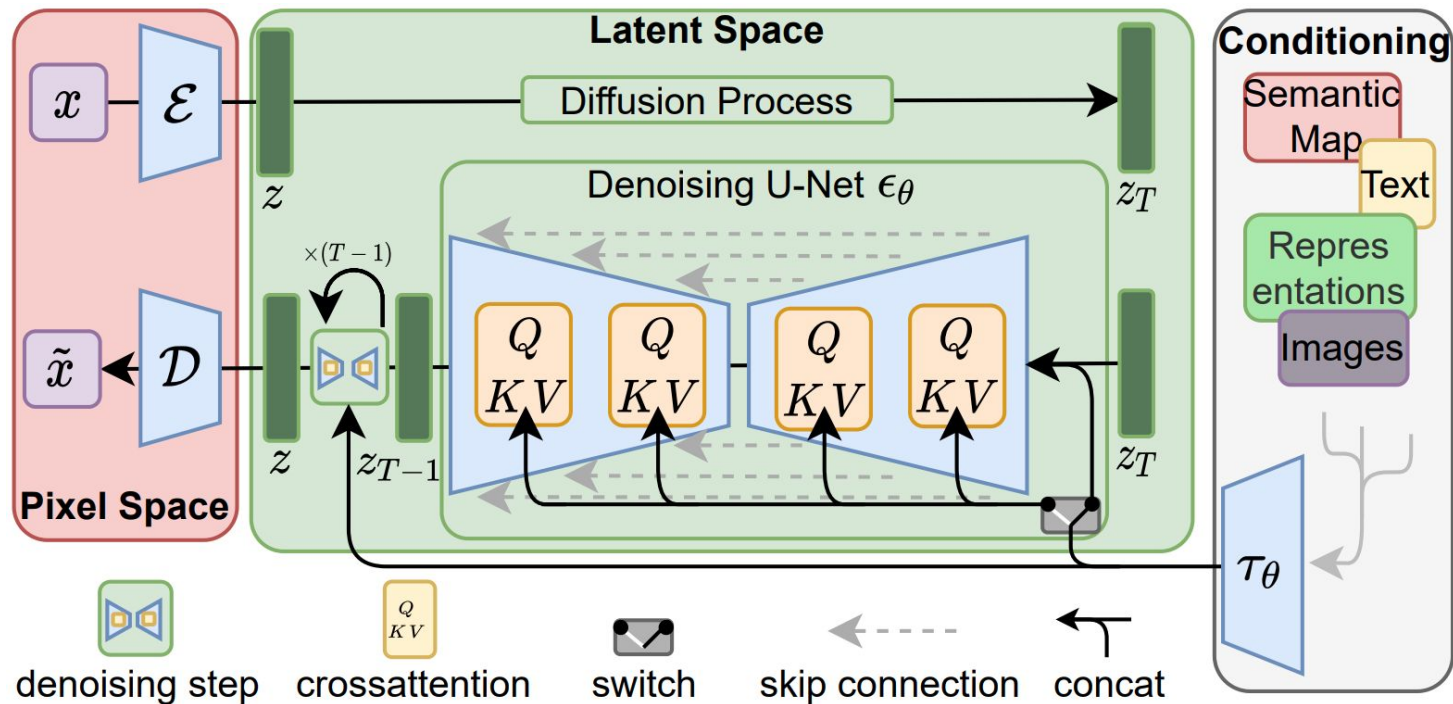
<https://c2pa.org/>

Reconstruction Attacks



- Black box
- Protection mechanism, model, and content independent
- Easy to compute
- Effective protection removal

Stable Diffusion Upscaler



Neutralizing Proactive Watermark

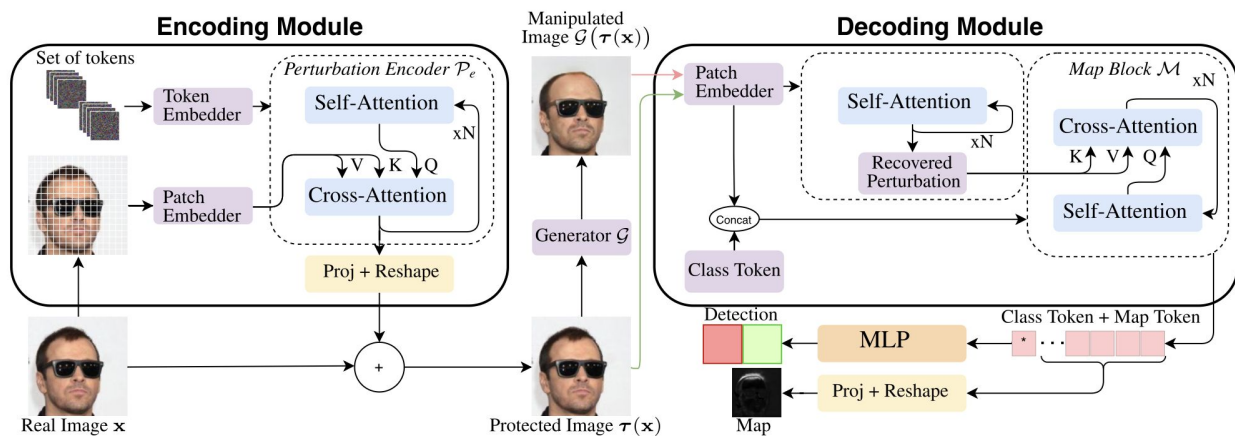
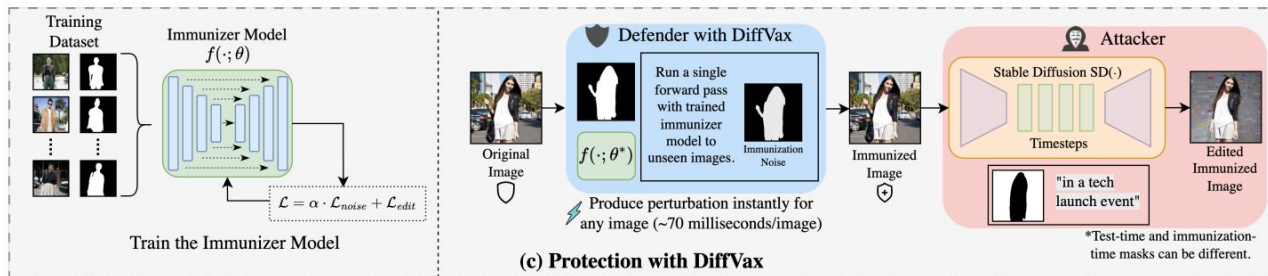


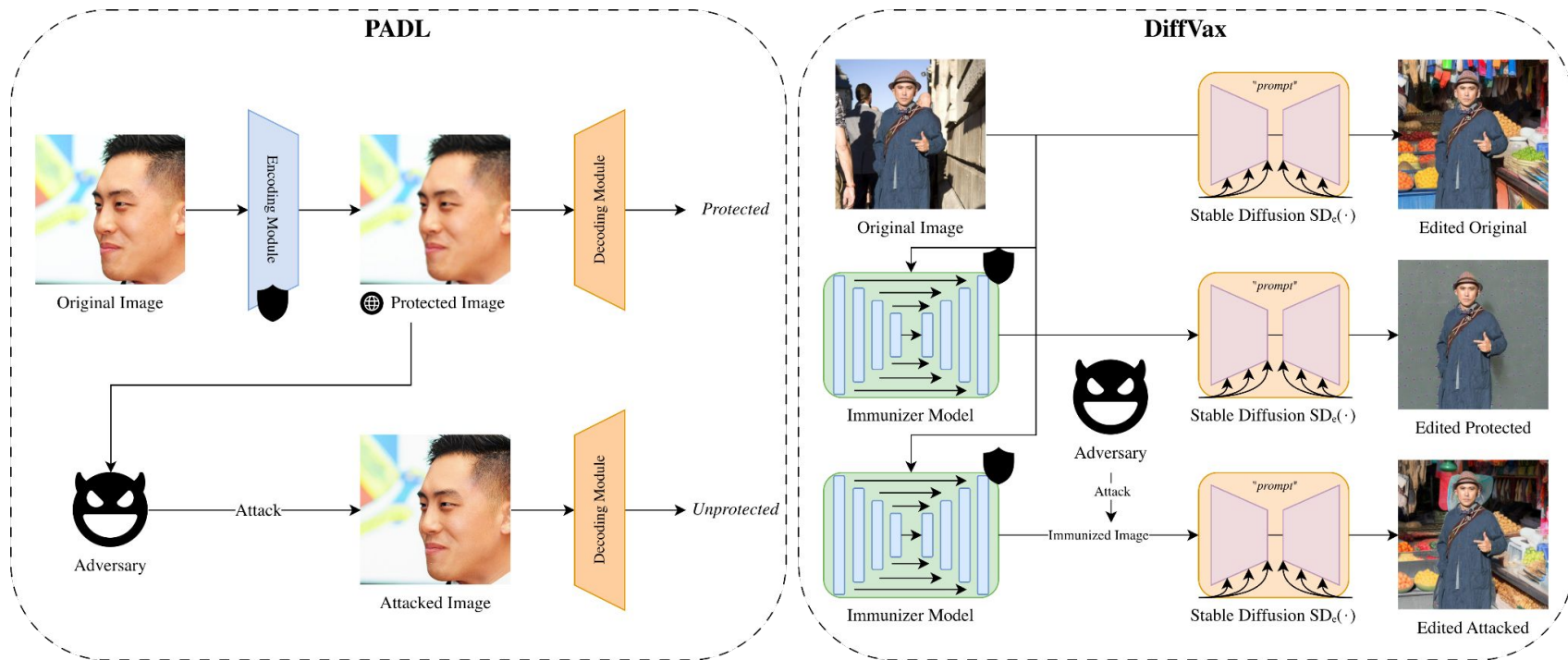
Image Specific Protection Against Manipulation, Bartolucci et al., *ICIAP* 2025

PADL, Bartolucci et al., *IEEE Access*, 2025

DiffVax, Ozden et al, *ICLR* 2026



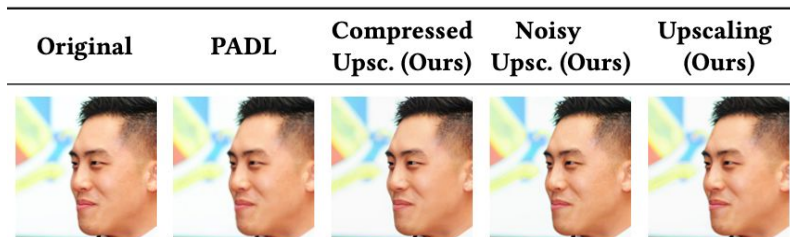
Neutralizing Proactive Defense



Neutralizing Proactive Defense

Key results:

- PADL: 100% attacked protected images are classified as unprotected.
- DiffVax: the attacks suppress the immunization noise.
- The attacked images do not degrade in terms of quality and similarity with the original images.
- Adversarial training does not improve robustness.



Takeaway Message

“In-pixel immunizations are not sufficient to protect digital content.”

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Thanks for your
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